

Acknowledgments

I was introduced to polarons half a century ago as a graduate student of Ted Holstein. Over time Ted and I became colleagues and close friends.

When I began to work with Ted we primarily viewed the study of polarons as an interesting academic pursuit. Many people in the condensed-matter community were unfamiliar with the concept of self-trapping and had never heard the term polaron. The idea of self-trapping received little or no mention in most introductions to solid-state physics.

It was uncertain that strong-coupling polarons were more than an oddity for which there were but a few examples. Over time reports of polaron effects grew. Evidence of polarons was reported in some transition-metal oxides, molecular solids and non-crystalline semiconductors. Experimentalists were bringing polarons to life.

Our simple models did not provide a satisfactory explanation of why these charge carriers formed polarons. It now seems that combinations of long-range and short-range electron–phonon interactions with the localizing effects of defects, impurities and disorder often trigger polaron formation.

This book offers an overview of what I have learned about polarons. Learning doesn't occur in a vacuum. I am grateful to the many people who have helped me.

I thank my collaborators for sharing work on a wide range of interesting problems. I am especially indebted to my experimental colleagues and collaborators for the data they present, the diversity of intricate materials they investigate and the questions that they and their work raise.

I was also lucky to have spent most of my career associated with Sandia National Laboratories. From my arrival in 1969 through my formal retirement I was continually able to devote myself fully to studying fundamental condensed-matter physics. My managers always supported my efforts with a sense of *noblesse oblige*. Science had not yet grown into a big enterprise.

I also appreciate the hospitality shown me by the Department of Physics and Astronomy of the University of New Mexico over the last 30 years. During this time I have collaborated with its faculty as well as its post-doctoral and graduate students. UNM has become my primary habitat since my retirement from Sandia.

For half a century I have received unflagging encouragement and support from my wife Shirley. She is remarkable in her continuing interest in learning about all of the science I study. I cannot imagine how my research and the writing of this book would have progressed without her steadfast and sensitive devotion.

Finally I wish to thank the editorial staff of Cambridge University Press for their continuous assistance, patience and understanding. It has taken quite a few years for the idea for this book to come to fruition.

